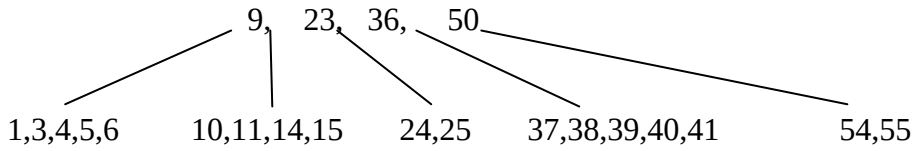
Q1.) Final 2015 (5+5+5+5 = 20 pts)
Consider the following B-tree with t=3. Show all your work for each part!

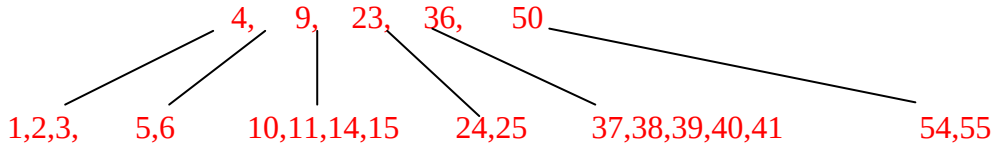


a) Insert 2
b) Insert 42
c) Delete 39
d) Delete 9

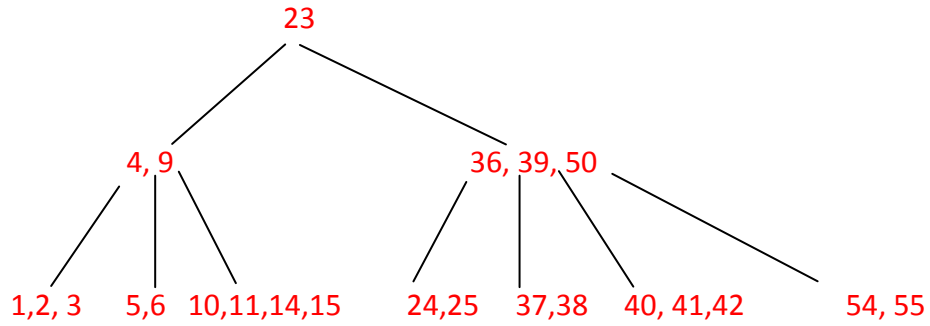
Since t=3, Maximum key = 2t-1=5

Minimum key = t-1=2

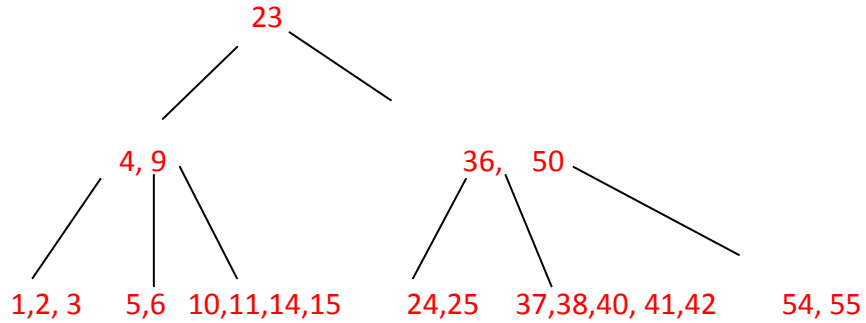
a)



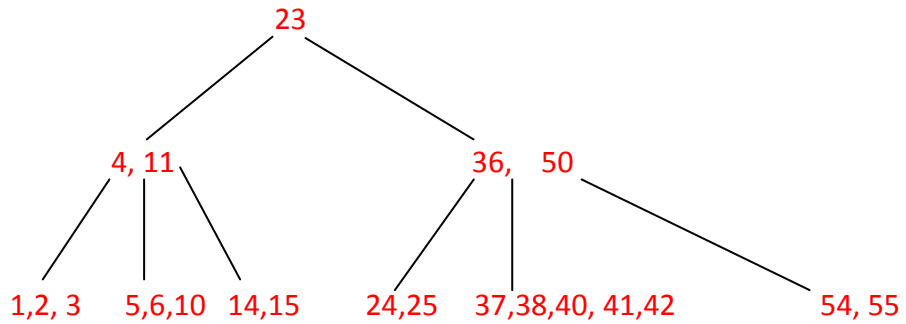
b)



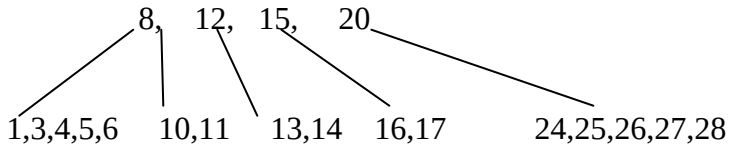
c)



d)

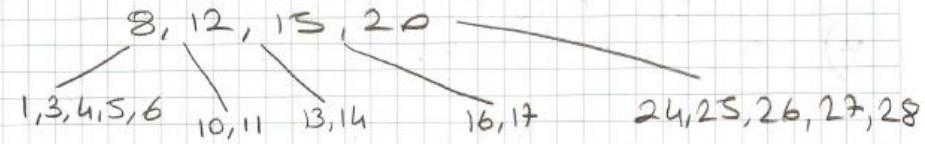


Q2.) Final 2016 (30 points) (HOMEWORK)
Consider the following B-tree with t=3.Show all your work for each part.



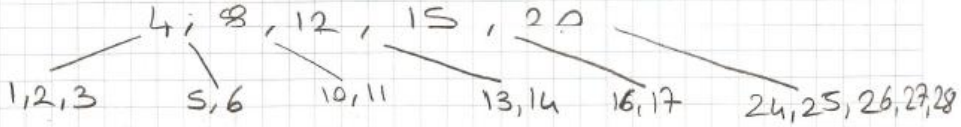
a) Insert 2 (**10 points**)
b) Insert 7(After inserting 2) (**10 points**)
c) Insert 30(After inserting 2 and 7) (**10 points**)

1. Consider the following tree=

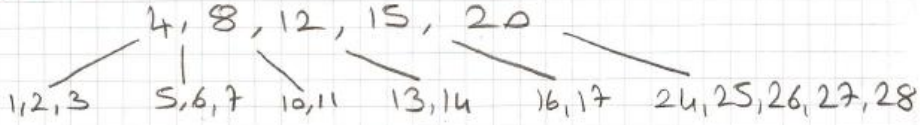


t=3
Max. key = 2t − 1 = 5
Min. key = t − 1 = 2

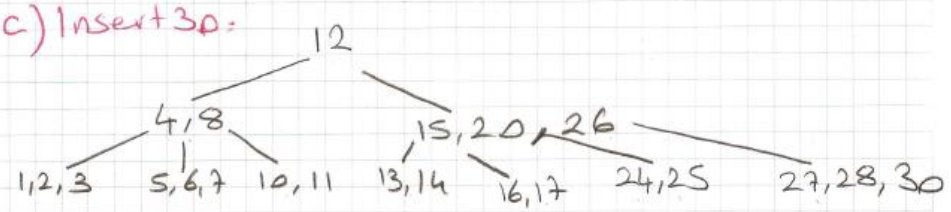
a) Insert 2=



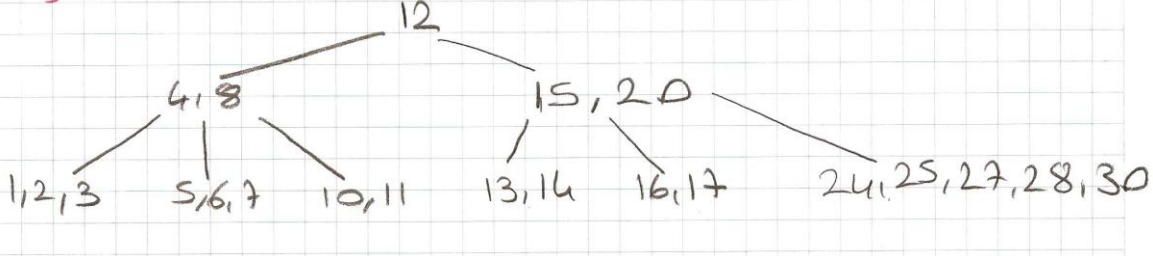
b) Insert 7:



c) Insert 30:

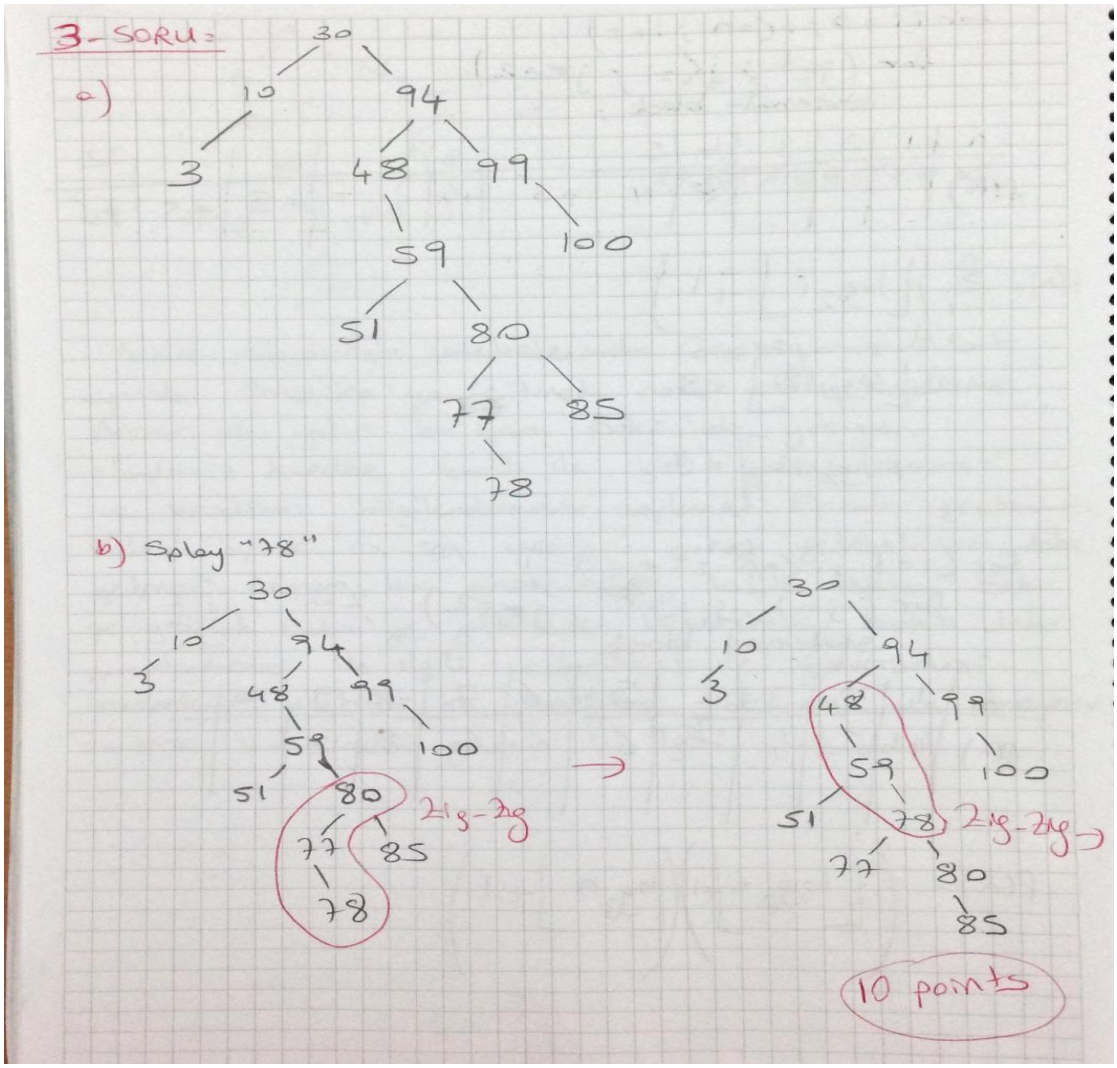


d) Delete 26:



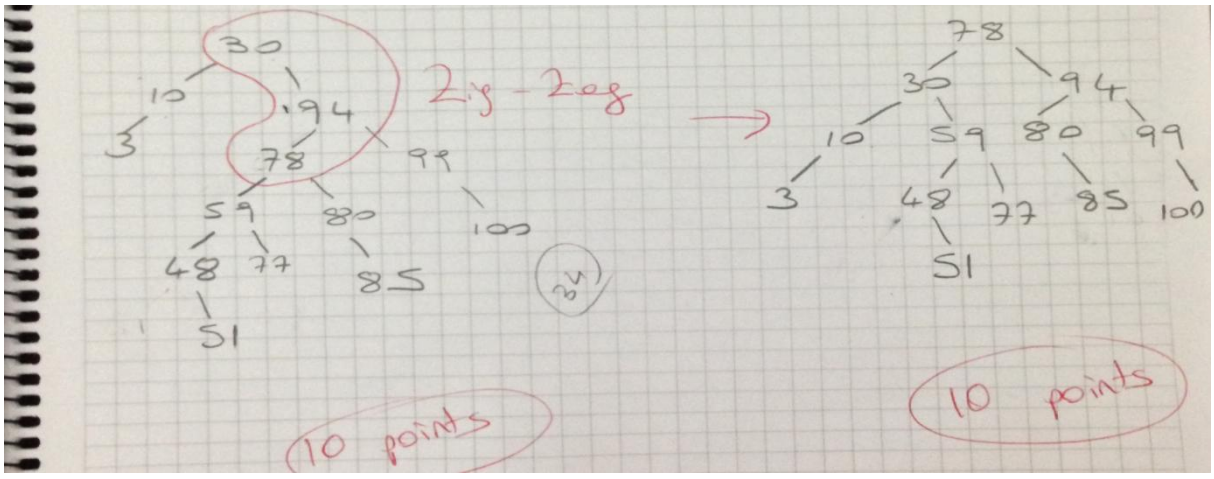
Q3.)Quiz-2016 Construct a BST for the following sequence. (**4 points**)
30, 10, 94, 3, 99, 48, 59, 100, 51, 80, 77, 85, 78

a) Splay the node with key ”78” (**30 points**)



Zig-Zag , double rotation, place the grandchild into the α-node, make the parent of the grandchild the left child and make the α-node the right child of the new root of the subtree.

Zig-zig, RR violation, single rotation, make the tree clockwise reverse

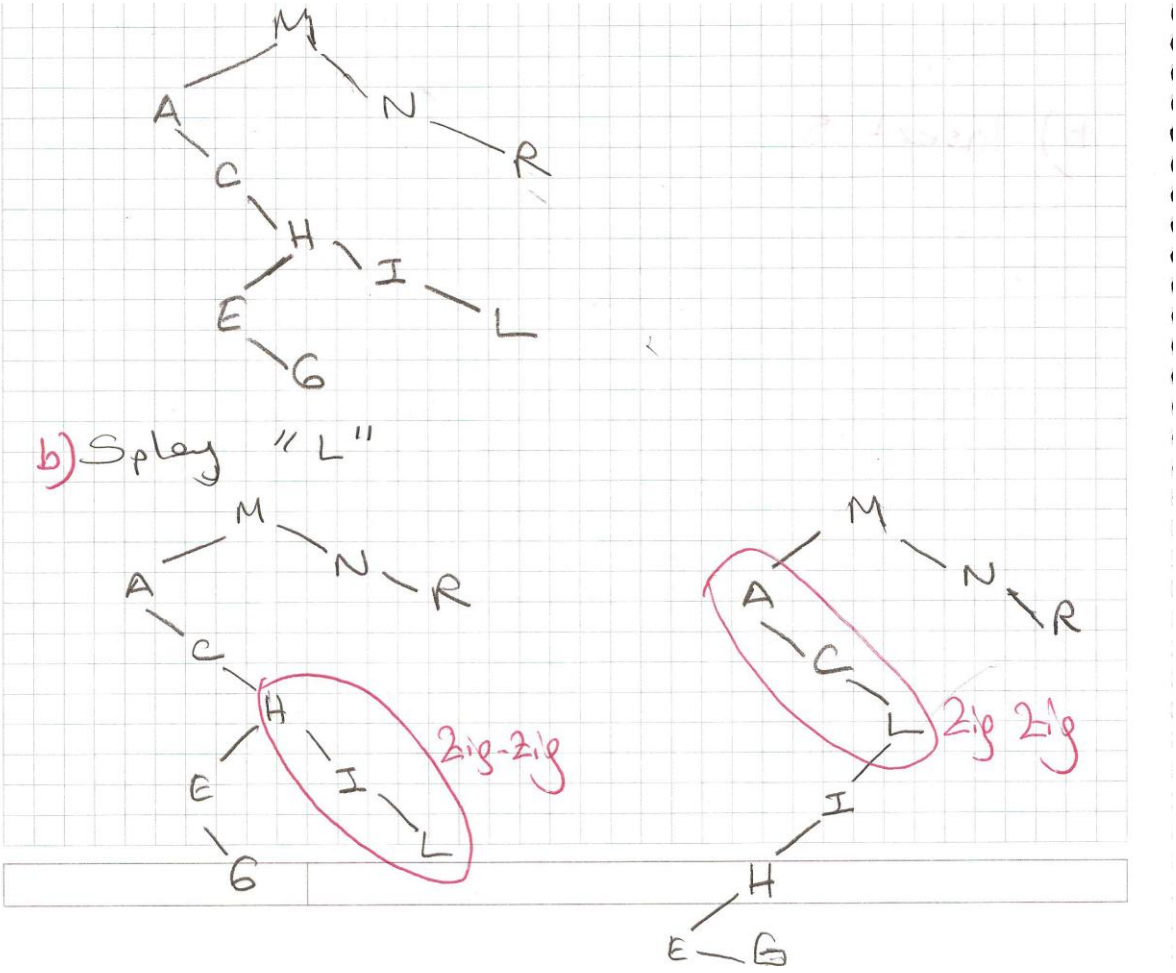


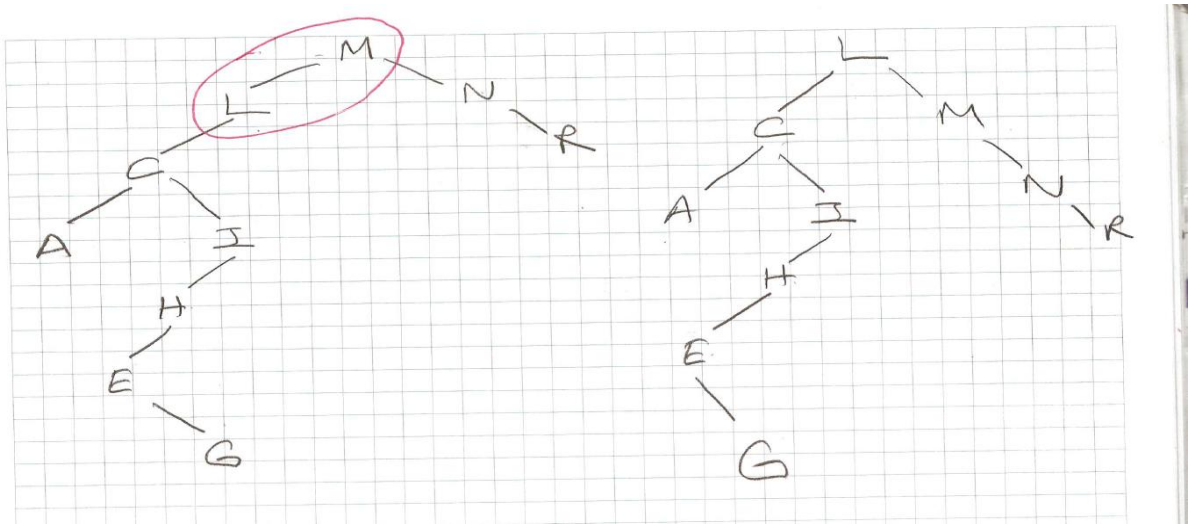
Zig-Zag , double rotation, place the grandchild into the α-node, make the parent of the grandchild the left child and make the α-node the right child of the new root of the subtree.

Q4.) **Quiz-2015 (30 points)**

a) Construct a BST from the letters "MACHINE LEARNING" where no duplicate letters are allowed.

b) Splay the node "L". (Show all your work with)





Q5. (18+8=26pts)

a) Insert the keys ***243, 175, 179,73, 273, 18 into*** the hash table of length ***m=17*** using open addressing with the hash function $h_1(k)=k \bmod m$. Use ***double hashing*** as the collision resolution strategy with $h_2(k)=R-(k \bmod R)$ where *R* is determined by following our in-class discussion.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
					243											
												175				
									179							
										73						
	273															
18																

243 mod 17=5 (empty)

175 mod 17=5 (full), 1 st collison

In general R is taken as the greatest prime number less than m.

R=13

Hash function using the double hashing strategy $h_{dh}(k,i)$:

$h(k,i)=[h(k)+i*(R-(k \bmod R))] \bmod m$

h(175,1)= [5+ 1*(13-(175 mod 13))] mod 17=[5+ 1*(13-6)] mod 17 =12 mod 17=12(empty)

179 mod 17=9(empty),

73 mod 17=5 (full)

h(73,1)= [5+ 1*(13-(73 mod 13))] mod 17=[5+ 1*(13-8)] mod 17 =10 mod 17=10(empty)

273 mod 17=1(empty)

18 mod 17=1 (full) ,1 st collison

h(18,1)= [1+ 1*(13-(18 mod 13))] mod 17=[1+ 1*(13-5)] mod 17 =9 mod 17=9 (full), 2 nd collison

h(18,2)=[1+ 2*(13-(18 mod 13))] mod 17=[1+ 2*(13-5)] mod 17 =17 mod 17=0 (empty)

b) (2+2 +2+2 pts)

i) What is the load factor?

ii) When would you decide to rehash?

iii) What is the cost for insert?

iv) What is the cost for a successful search?

i) 6/17

ii) >1/2

iii)

Load factor: a=6/17

Cost for insert: $C_{ins}=1/(1-a) = 1/(1-6/17)=1/(11/17)=17/11=1.55$

iv)Cost for successful search: $C_{search}=(1/a)\ln(1/(1-a)) = (17/6)*\ln(17/11)=1.25$

Q6. (2015) (15+10=25pts)

a) Insert the keys ***211, 376, 259,137, 29 into*** the hash table of length ***m=13*** using open addressing with the hash function $h_1(k)=k \bmod m$. Use ***double hashing*** as the collision resolution strategy with $h_2(k)=R-(k \bmod R)$.

0	1	2	3	4	5	6	7	8	9	10	11	12
			211									
												376
				259								
							137					
											29	

211 mod 13=3 (empty)

376 mod 13=12 (empty)

259 mod 13=12(full), 1 st collison

In general R is taken as the greatest prime number less than m.

R=11

Hash function using the double hashing strategy $h_{dh}(k,i)$:

$h(k,i)=[h(k)+i*(R-(k \bmod R))] \bmod m$

h(259,1)= [12+ 1*(11-(259 mod 11))] mod 13

h(259,1)=[12+ 1*(11-6)] mod 13 = 17 mod 13=4 (empty),

137 mod 13=7 (empty)

29 mod 13=3 (full)

h(29,1)= [3+ 1*(11-(29 mod 11))] mod 13=)=[3+ 1*(11-7)] mod 13) =7 mod 13=7 8 (full)

h(29,2)=[3+ 2*4)] mod 13 = 11 mod 13=11 (empty),

b) (5+5 pts)

v) What is the load factor?

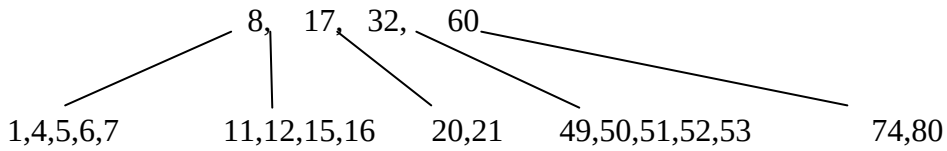
vi) When would you decide to rehash?

LOAD FACTOR= number of occupied cells/ number of total cells =5/13

Load factor>1/2

Q7. (4+4+4+4 = 16 pts)

Consider the following B-tree with t=3. Show all your work for each part!



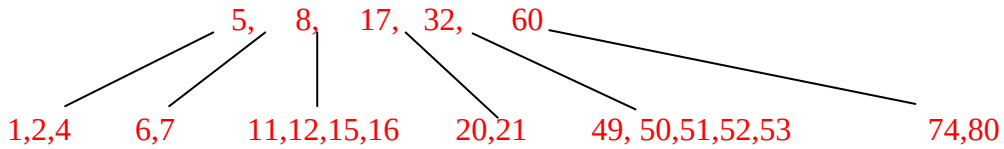
e) Insert 2
f) Insert 35
g) Delete 8
h) Delete 60

Since t=3

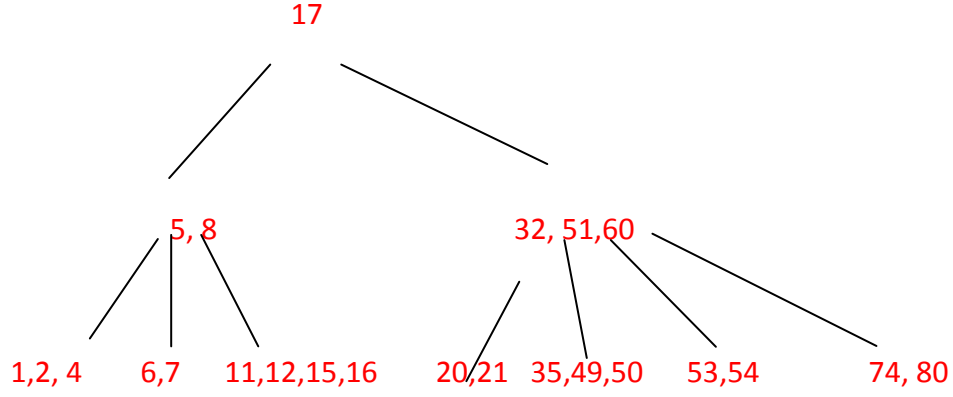
Maximum key = 2t-1=5

Minimum key = t-1=2

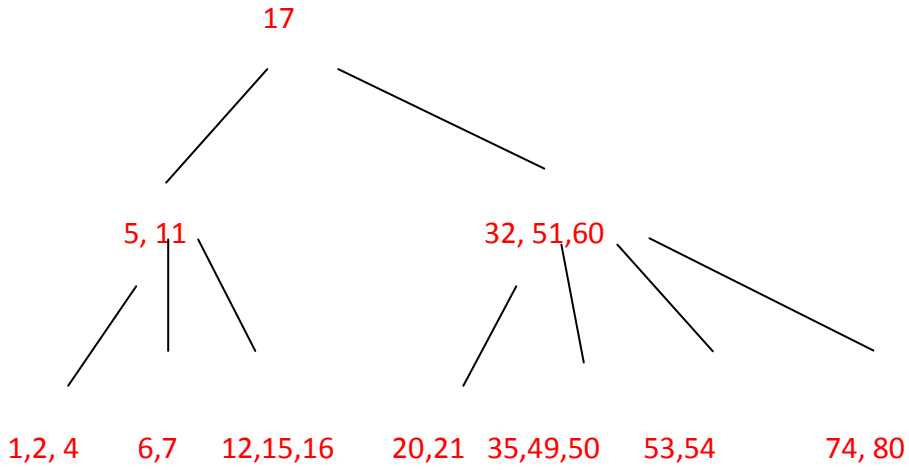
a)



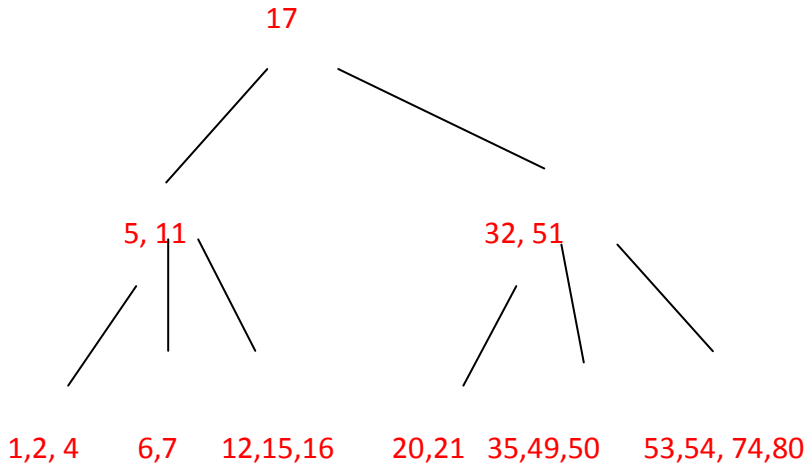
b)



c)



d)



Q8. (5+10 points)

c) Construct a BST from the letters ”KAHRAMAN FETHİ ASKER” where no duplicate letters are allowed.

d) Splay the node ”E”. (Show all your work)

Q9. FİNAL 2014 (10+15=25pts)

Assume an AVL tree topology of height 6 (from depth level 0 to 5) with a minimum number of nodes.

a) How many such binary tree topologies (not only AVL tree topologies but also those) are there provided that the number of nodes at each depth level remains constant? Show one sample AVL tree topology.

b) How many binary tree topologies (NOT AVL tree topologies) are there with the same total search time or internal path length?